

BUSAN, Korea, Republic of

WMO: 471590

Lat: 35.1047N

Lon: 129.0319E

Elev: 233

StdP: 14.57

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 22.7	(c) 25.9	(d) -6.9	(e) 3.8	(f) 28.7	(g) -3.2	(h) 4.7	(i) 30.5	(j) 22.6	(k) 36.4	(l) 20.5	(m) 35.5	(n) 9.4	(o) 340	(p) 0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 9.0	(c) 88.7	(d) 78.1	(e) 86.7	(f) 77.4	(g) 84.7	(h) 76.5	(i) 79.8	(j) 85.5	(k) 78.8	(l) 84.3	(m) 77.8	(n) 83.1	(o) 9.6	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.2	147.8	83.1	77.2	142.6	82.5	76.1	137.4	81.7	43.6	85.9	42.5	84.3	41.5	83.2	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.1	18.2	15.9		DB	17.2	91.6	4.4	2.9	14.1	93.7	11.5	95.4	9.0	97.1	5.7	99.2
				WB	13.2	81.1	3.6	2.1	10.7	82.6	8.6	83.9	6.6	85.1	4.0	86.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.6	38.8	42.2	48.7	57.2	64.7	70.3	76.7	79.9	73.3	65.2	54.1	42.9
	DBStd	14.64	6.43	6.53	5.86	4.90	4.08	3.20	4.61	4.07	3.97	4.80	6.64	7.19
	HDD50	948	348	227	94	6	0	0	0	0	0	1	36	236
	HDD65	3317	811	639	504	236	56	3	0	0	0	55	328	685
	CDD50	4451	2	9	55	222	455	608	827	927	700	471	160	15
	CDD65	1346	0	0	0	2	47	160	362	462	250	61	2	0
	CDH74	8588	0	0	0	3	86	344	2597	4324	1143	91	0	0
	CDH80	2087	0	0	0	0	3	12	612	1302	157	1	0	0
Wind	WSAvg	7.7	7.9	7.8	8.5	8.5	7.8	7.1	8.1	7.8	7.5	6.8	6.9	7.4
Precipitation	PrecAvg	59.0	1.2	1.8	3.3	5.7	5.9	7.9	11.0	9.1	6.9	2.8	2.3	1.1
	PrecMax	94.4	4.9	5.0	8.7	15.2	16.5	36.9	34.9	27.6	16.4	9.8	10.5	6.4
	PrecMin	35.5	0.0	0.0	0.2	0.5	0.0	0.7	2.1	0.7	0.7	0.0	0.0	0.0
	PrecStd	15.7	1.0	1.2	1.9	3.0	3.5	5.9	6.5	6.8	4.7	2.2	2.1	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.3	60.8	65.6	72.9	79.3	81.6	90.0	92.2	86.5	78.8	71.8	61.7
		MCWB	50.4	51.8	53.5	58.4	62.9	69.6	78.6	78.9	76.6	66.9	61.0	53.3
	2%	DB	53.1	57.3	62.5	69.8	76.3	79.4	87.6	89.7	83.7	76.4	68.3	58.4
		MCWB	44.7	49.4	52.4	57.2	62.0	68.8	77.9	78.3	74.2	65.5	58.8	50.4
	5%	DB	50.4	54.4	59.9	67.3	74.1	77.6	85.5	87.9	81.6	74.5	65.8	55.6
		MCWB	42.6	46.3	51.1	56.3	61.9	68.0	77.3	77.8	72.4	64.6	57.5	47.3
	10%	DB	47.8	51.7	57.7	65.0	71.9	75.9	83.5	86.2	79.9	72.6	63.4	53.0
		MCWB	40.3	43.8	49.8	55.4	61.0	67.4	76.5	77.2	71.3	63.2	55.8	45.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.2	56.6	58.5	64.3	68.5	74.7	81.4	81.8	78.9	72.2	64.6	56.0
		MCDB	55.4	59.0	61.7	67.1	72.1	77.1	86.4	87.9	84.3	74.9	68.1	58.5
	2%	WB	47.2	51.9	56.0	61.4	66.6	72.8	79.6	80.1	76.9	69.2	61.8	52.0
		MCDB	51.0	55.3	59.5	65.3	71.1	75.2	84.8	86.3	81.3	74.0	66.0	57.1
	5%	WB	44.2	47.4	53.5	59.6	65.1	71.4	78.5	79.2	75.1	66.9	59.2	48.9
		MCDB	48.8	53.0	58.1	64.1	70.2	74.4	83.5	85.0	79.9	72.1	64.8	54.6
	10%	WB	41.7	44.7	50.8	57.7	63.7	70.0	77.3	78.3	73.0	64.7	56.8	46.2
		MCDB	46.6	50.5	56.4	62.8	69.1	73.9	82.0	84.0	78.2	70.8	62.3	52.3
Mean Daily Temperature Range	5% DB	MDBR	13.0	13.4	13.1	12.4	11.5	9.5	8.2	9.0	10.2	12.4	12.8	13.0
		MCDBR	14.0	14.6	14.5	14.8	14.4	11.8	10.4	11.1	11.6	13.4	13.5	13.9
		MCWBR	10.4	10.5	10.1	9.6	8.8	6.8	4.9	5.2	6.4	7.7	9.3	10.6
	5% WB	MCDBR	11.9	12.6	11.7	11.4	11.3	8.1	9.3	9.8	9.5	10.6	11.7	12.9
MCWBR		10.7	11.2	11.1	9.1	8.2	5.7	4.9	5.2	6.1	7.8	9.8	11.1	
Clear-Sky Solar Irradiance	taub		0.406	0.456	0.528	0.558	0.570	0.607	0.582	0.539	0.490	0.455	0.424	0.399
	taud		2.099	1.998	1.839	1.821	1.846	1.833	1.958	2.075	2.156	2.132	2.131	2.132
	Ebn at Noon		242	243	235	237	236	226	232	239	245	241	234	235
	Edh at Noon		39	49	62	66	66	66	58	51	45	42	38	36
All-Sky Solar Radiation	RadAvg		843	1043	1362	1581	1733	1528	1467	1492	1239	1139	870	772
	RadStd		68	116	125	124	186	139	225	180	131	109	103	46

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A